

DBS302 - NoSQL Database Management

Practical 2B: Real-Time Analytics with Bitmaps and HyperLogLog

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1. Aim

The aim of this project is to use Redis bitmaps to implement user analytics. This will be achieved by utilizing Redis bitmaps to track user activities. Additionally, Redis' HyperLogLog data structure will be employed to count the number of unique users.

2. Objectives

- The objectives of the project are as follows:
- Conceptually explain Redis bitmaps and HyperLogLog
- Use Redis bitmaps to implement daily active user tracking
- Use Redis' HyperLogLog data structure to implement daily unique visitor counting
- Develop an analytics service that provides a high-level interface to access user analytics
- Complete a series of exercises involving weekly UV merging, stickiness ratio calculation, and creating a CLI script.

3. Conceptual Overview

3.1 Bitmaps

A bitmap uses bits in a string to represent binary states (active/inactive), typically one bit per user per day.

Key Commands:

- SETBIT key offset value - Set a bit at a specific position
- GETBIT key offset - Get the value of a bit
- BITCOUNT key - Count number of bits set to 1

3.2 HyperLogLog (HLL)

HyperLogLog is a probabilistic data structure that estimates the number of unique elements with very low memory usage and an acceptable error rate (~0.81%).

Key Commands:

- PFADD key element [element ...] - Add elements to HyperLogLog
- PFCOUNT key - Get approximate cardinality
- PFMERGE destkey sourcekey [sourcekey ...] - Merge multiple HyperLogLogs

4. Bitmap Implementation - Daily Active Users (DAU)

4.1 Redis CLI Demonstration

```
macbookairm4chip@Tandin-0mu redis-practicals % redis-cli
127.0.0.1:6379> SETBIT analytics:active:2026-03-17 1 1
(integer) 0
127.0.0.1:6379> SETBIT analytics:active:2026-03-17 42 1
(integer) 0
127.0.0.1:6379> SETBIT analytics:active:2026-03-17 100 1
(integer) 0
127.0.0.1:6379> GETBIT analytics:active:2026-03-17 42
(integer) 1
127.0.0.1:6379> BITCOUNT analytics:active:2026-03-17
(integer) 3
127.0.0.1:6379> 
```

The following commands were executed to demonstrate bitmap operations for tracking daily active users:

Command	Output	Purpose
SETBIT analytics:active:2026-03-17 1 1	(integer) 0	Mark user 1 as active
SETBIT analytics:active:2026-03-17 42 1	(integer) 0	Mark user 42 as active
SETBIT analytics:active:2026-03-17 100 1	(integer) 0	Mark user 100 as active
GETBIT analytics:active:2026-03-17 42	(integer) 1	Check if user 42 was active :True
BITCOUNT analytics:active:2026-03-17	(integer) 3	Count total active users : 3

5. HyperLogLog Implementation - Unique Visitors (UV)

5.1 Redis CLI Demonstration

```
127.0.0.1:6379> PFADD analytics:uv:2026-03-17 user1 user2 user3
(integer) 1
127.0.0.1:6379> PFADD analytics:uv:2026-03-17 user2 user3 user4
(integer) 1
127.0.0.1:6379> PFCOUNT analytics:uv:2026-03-17
(integer) 4
127.0.0.1:6379> █
```

6. Python Analytics Service Implementation

```
• (venv) macbookairm4chip@Tandin-0mu redis-practicals % python analytics.py
Simulating activity for 2026-03-17...

Is user 42 active?
-> True

Daily Active Users (DAU): 3
Unique Visitors (UV) [approx]: 4
```

A `RealtimeAnalytics` class was implemented in Python with the following methods:

Method	Description	Redis Command
<code>mark_user_active(date, user_id)</code>	Mark user as active for a day	SETBIT
<code>is_user_active(date, user_id)</code>	Check if user was active	GETBIT
<code>count_daily_active_users(date)</code>	Count daily active users	BITCOUNT
<code>add_visit(date, user_id)</code>	Record a visit	PFADD
<code>count_unique_visitors(date)</code>	Count unique visitors	PFCOUNT

7. Exercise 1: Merge Multiple HyperLogLogs (Weekly UV)

7.1 CLI Demonstration with PFMERGE

```
(venv) macbookairm4chip@Tandin-0mu redis-practicals % redis-cli
127.0.0.1:6379> DEL analytics:uv:2026-03-15
(integer) 0
127.0.0.1:6379> DEL analytics:uv:2026-03-16
(integer) 0
127.0.0.1:6379> DEL analytics:uv:2026-03-17
(integer) 1
127.0.0.1:6379> DEL analytics:uv:week-2026-03-15
(integer) 0
127.0.0.1:6379> PFADD analytics:uv:2026-03-15 user1 user2 user3
(integer) 1
127.0.0.1:6379> PFADD analytics:uv:2026-03-16 user2 user3 user4 user5
(integer) 1
127.0.0.1:6379> PFADD analytics:uv:2026-03-17 user3 user4 user5 user6
(integer) 1
127.0.0.1:6379> PFCOUNT analytics:uv:2026-03-15
(integer) 3
127.0.0.1:6379> PFCOUNT analytics:uv:2026-03-16
(integer) 4
127.0.0.1:6379> PFCOUNT analytics:uv:2026-03-17
(integer) 4
127.0.0.1:6379> PFMERGE analytics:uv:week-2026-03-15 analytics:uv:2026-03-15 analytics:uv:2026-03-16 analytics:uv:2026-03-17
OK
127.0.0.1:6379> PFCOUNT analytics:uv:week-2026-03-15
(integer) 6
127.0.0.1:6379>
```

Analysis: The merged weekly count correctly identified 6 unique users across the three days.

7.2 Python Implementation

```
(venv) macbookairm4chip@Tandin-0mu redis-practicals % python analytics.py

Simulating activity for 2026-03-17...

Is user 42 active?
-> True

Daily Active Users (DAU): 3
Unique Visitors (UV) [approx]: 4

=====
EXERCISE 1: Weekly Unique Visitors (PFMERGE)
=====

Individual Day Counts:
  March 15: 3 unique visitors
  March 16: 4 unique visitors
  March 17: 6 unique visitors

Weekly Unique Visitors (March 15-17): 6
(Should be 6 unique users: user1, user2, user3, user4, user5, user6)
(venv) macbookairm4chip@Tandin-0mu redis-practicals %
```

8. Exercise 2: Stickiness Ratio (DAU / MAU)

8.1 CLI Demonstration - MAU Concept

```
(venv) macbookairm4chip@Tandin-Omu redis-practicals % redis-cli
127.0.0.1:6379> FLUSHDB
OK
127.0.0.1:6379> PFADD analytics:uv:2026-02-15 user1 user2 user3
(integer) 1
127.0.0.1:6379> PFADD analytics:uv:2026-02-16 user1 user2 user3 user4
(integer) 1
127.0.0.1:6379> PFADD analytics:uv:2026-02-17 user1 user2 user3 user4 user5
(integer) 1
127.0.0.1:6379> PFADD analytics:uv:2026-03-15 user1 user2 user3
(integer) 1
127.0.0.1:6379> PFADD analytics:uv:2026-03-16 user2 user3 user4 user5
(integer) 1
127.0.0.1:6379> PFADD analytics:uv:2026-03-17 user3 user4 user5 user6
(integer) 1
127.0.0.1:6379> PFCOUNT analytics:uv:2026-03-17
(integer) 4
127.0.0.1:6379> PFMERGE analytics:uv:february-2026 analytics:uv:2026-02-15 analytics:uv:2026-02-16 analytics:uv:2026-02-17
OK
127.0.0.1:6379> PFCOUNT analytics:uv:february-2026
(integer) 5
127.0.0.1:6379> █
```

Monthly Active Users (MAU) was calculated by merging daily HyperLogLogs.

8.2 Python Implementation

```
(venv) macbookairm4chip@Tandin-Omu redis-practicals % python analytics.py
Simulating activity for 2026-03-17...

Is user 42 active?
-> True

Daily Active Users (DAU): 3
Unique Visitors (UV) [approx]: 4

=====
EXERCISE 1: Weekly Unique Visitors (PFMERGE)
=====

Individual Day Counts:
  March 15: 3 unique visitors
  March 16: 4 unique visitors
  March 17: 4 unique visitors

Weekly Unique Visitors (March 15-17): 6
(Should be 6 unique users: user1, user2, user3, user4, user5, user6)

=====
EXERCISE 2: Stickiness Ratio (DAU / MAU)
=====

Stickiness Analysis for March 17:
  Daily Active Users (DAU): 4
  Monthly Active Users (MAU): 9
  Stickiness Ratio: 44.44%

  (Stickiness ratio shows how engaged users are -
   higher percentage means users visit frequently)
(venv) macbookairm4chip@Tandin-Omu redis-practicals % █
```

Interpretation: A stickiness ratio of 44.44% indicates that on average, active users visit the platform about 13-14 days per month (44.44% of 30 days).

9. Exercise 3: CLI Script for DAU and UV

9.1 Redis CLI Query Concept

```
(venv) macbookairm4chip@Tandin-0mu redis-practicals % redis-cli
127.0.0.1:6379> BITCOUNT analytics:dau:2026-03-17
(integer) 3
127.0.0.1:6379> PFCOUNT analytics:uv:2026-03-17
(integer) 4
127.0.0.1:6379>
```

9.2 Python CLI Script

```
(venv) macbookairm4chip@Tandin-0mu redis-practicals % python analytics_cli.py 2026-03-17

=====
Analytics Report for 2026-03-17
=====
Daily Active Users (DAU): 3
Unique Visitors (UV):    6
=====

Note: DAU is exact count from bitmap
UV is approximate from HyperLogLog (error rate ~0.81%)
(venv) macbookairm4chip@Tandin-0mu redis-practicals % python analytics_cli.py 2026-03-15

=====
Analytics Report for 2026-03-15
=====
Daily Active Users (DAU): 0
Unique Visitors (UV):    3
=====

(venv) macbookairm4chip@Tandin-0mu redis-practicals % python analytics_cli.py 2026-03-16

=====
Analytics Report for 2026-03-16
=====
Daily Active Users (DAU): 0
Unique Visitors (UV):    4
=====

(venv) macbookairm4chip@Tandin-0mu redis-practicals %
```

A command-line script `analytics_cli.py` was created that accepts a date as an argument and displays DAU and UV.

11. Key Learnings

Bitmaps

- Good for tracking binary states like active/inactive per user per day
- BITCOUNT is good for aggregation of active users
- Memory usage is low, 1 bit per user

HyperLogLog

- Good for getting approximate unique user counts with low memory usage
- The error rate is approximately 0.81%
- PFMERGE is good for combining multiple days to get weekly/monthly analytics

Stickiness Ratio

- Formula: $(DAU / MAU) * 100$
- The higher the percentage, the better the user engagement
- Good for tracking user adoption of a product